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# DETAIL SPECIFICATION

# PRINTED WIRING ASSEMBLY, MUX/ADCON

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the Multiplex/Analog to Digital Converter (Mux/ADCON) Printed Wiring Assembly, referred to herein as the PWA. This is one of the six printed wiring assemblies that comprise the Simplified Test Equipment/Internal Combustion Engine-Repregrammable (STE/ICE-R) Vehicle Test Meter (VTM) (see Drawing 12259265).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5998

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12258777 - Printed Wiring Assembly, MUX/ADCON.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies.



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(Application for copies should be addressed to the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample PWA shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Design, materials, and manufacturing process selection shall be as specified herein and in applicable referenced specifications, standards and drawings. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All metallic parts shall be made from corrosion resistant metals or treated with corrosion-resistant materials. Asbestos and cadmium materials shall not be used in any form in any part of the PWA. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each PWA shall provide the following functional, operational, and performance capabilities.

3.3.1 Input power. The PWA shall require not more than the current specified in table I.

TABLE I. Input power.

| Source | Voltage | Current |
|---|---|---|
| P1-5, 8 | +5 Vdc 1/ | 0.130 A 2/ |
| P1-13, 14 | +15 Vdc | 0.040 A |
| P1-19, 20 | -15 Vdc | 0.040 A |
| P1-7, 8 | +8 Vdc | 0.004 A |
| P1-17, 18 | -8 Vdc | 0.004 A |

1/ Vdc = Volts direct current
2/ A = ampere



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3.3.2 Complementary metal oxide semiconductor (CMOS) logic levels. Unless otherwise specified (see 6.2), CMOS logic levels accepted by and output from the PWA shall be as follows:

Input High (Logic Level 1) = 3.5 Vdc minimum (min)
Input Low (Logic Level 0) = 1.5 Vdc maximum (max.)
Output High (Logic Level 1) = 4.5 Vdc min
Output Low (Logic Level 0) = 0.5 Vdc max.

3.3.3 Input/output logic. (NOTE: A "/" following a capitalized logic name denotes logic negation.)

3.3.3.1 Output clocks. The signal at U7-10 shall be determined by the signals at P1-29 (IO9/1) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 1. The signal at U7-4 shall be determined by the signals at P1-30 (IOA/2) and P1-37 (OEN-Q/) and shall conform to the waveforms shown in figure 2.

3.3.3.2 Output latches. Logic levels at P1-21 through P1-26 (BUS 0 through BUS 5) shall be latched by U1 and appear at U1 pins 2, 5, 7, 10, 15 and 12, respectively, when clocked by U7-10 as shown in figure 3. Logic levels at P1-21 through P1-26 (BUS 0 through BUS 5) shall be latched by U2 and appear at U2 pins 15, 12, 10, 7, 2, and 5, respectively, when clocked by U7-4 as shown in figure 4.

3.3.3.3 Voltage level shifters. Logic levels present at U3, U4, and U5 pins 3, 6, 10, and 14 shall be shifted as shown below and shall be present at U3, U4, and U5 pins 4, 5, 13, and 11, respectively, within 600 nanoseconds (ns) of their application to the input.

Output High (Logic Level 1) = [(voltages at U5 pin 16) - 0.30 Vdc] minimum
Output Low (Logic Level 0) = 0.5 Vdc maximum

3.3.3.4 Input clocks. The signal at U7-3 shall be determined by the signals present at P1-29 (I09/1) and P1-36 (MRD/) and shall conform to the waveforms shown in figure 5. The signal at U7-11 shall be determined by the signals present at P1-30 (I0A/2) and P1-36 (MRD/) and shall conform to the waveforms shown in figure 6.

3.3.3.5 Input buffers. Logic levels present at U16 pins 12, 14, and U24 pins 2, 4, 14, 12, 10, and 6 shall be present on P1-28 through P1-21 (BUS 7 through BUS 0) when the signal at U7-3 is active as shown in figure 7. Logic levels present at U6 pins 6, 2, 4, 10, and U16 pins 6, 4, 2, and 10 shall be present on P1-28 through P1-21 when the signal at U7-11 is active as shown in figure 8. The signal present at P1-40 shall be present at U6-11 within 90 ns of its application at P1-40 regardless of any other signal.



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3.3.4 Internal logic levels. Logic levels accepted by U8, U9, U13, U14, U18, U19, U20, U21, U22, U23, and U26 shall be:

Input High (Logic Level 1) = 5.6 Vdc min
Input Low (Logic Level 0) = 2.4 Vdc max.

3.3.5 Input channel Mux select. The inputs specified in table II shall be routed to U10 pins 1 (+) and 3 (-) when the corresponding bit pattern from table II is applied to BUS 0 through BUS 5 in accordance with the waveforms shown in figure 3.

3.3.6 Input channel attenuation. The ratio of the magnitude of the input voltage to the magnitude of the voltage across U10 pins 1 and 3 shall be as stated by the attenuation ratio in table II for the channel selected in accordance with the requirements of paragraph 3.3.5. The attenuation accuracy shall be as stated in table II for the selected channel (if applicable).

TABLE II. Input channel attenuation and select.

| (+) Input | (-) Input | Attenuation | | BUS | | | | | | Input voltage range |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Ratio | Accuracy | 5 | 4 | 3 | 2 | 1 | 0 | |
| P1-98 | P1-99 | 1:1 | - | 0 | 0 | 0 | 0 | 0 | 0 | ± 5 V |
| P1-96 | P1-93 | 1:1 | - | 0 | 0 | 0 | 0 | 0 | 1 | ± 5 V |
| P1-94 | P1-97 | 1:1 | - | 0 | 0 | 0 | 0 | 1 | 0 | ± 5 V |
| P1-100 | P1-95 | 1:1 | - | 0 | 0 | 0 | 0 | 1 | 1 | ± 5 V |
| P1-90 | P1-91 | 1:1 | - | 0 | 0 | 0 | 1 | 0 | 0 | ± 5 V |
| P1-88 | P1-85 | 1:1 | - | 0 | 0 | 0 | 1 | 0 | 1 | ± 5 V |
| P1-86 | P1-89 | 1:1 | - | 0 | 0 | 0 | 1 | 1 | 0 | ± 5 V |
| P1-92 | P1-87 | 1:1 | - | 0 | 0 | 0 | 1 | 1 | 1 | ± 5 V |
| P1-82 | P1-83 | 1:1 | 1/ | 0 | 0 | 1 | 0 | 0 | 0 | ± 5 V |
| P1-80 | P1-77 | 1:1 | - | 0 | 0 | 1 | 0 | 0 | 1 | ± 5 V |
| P1-78 | P1-81 | 1:1 | - | 0 | 0 | 1 | 0 | 1 | 0 | ± 5 V |
| P1-84 | P1-79 | 1:1 | - | 0 | 0 | 1 | 0 | 1 | 1 | ± 5 V |
| P1-61 | P1-59 | 1:1 | - | 0 | 0 | 1 | 1 | 0 | 0 | ± 5 V |
| P1-55 | P1-53 | 10:1 | 0.2% | 0 | 0 | 1 | 1 | 1 | 0 | ± 50 V |
| P1-51 | P1-49 | 7:1 | 0.1% | 0 | 0 | 1 | 1 | 1 | 1 | ± 35 V |
| P1-47 | P1-49 | 7:1 | 0.1% | 0 | 1 | X | X | 0 | 0 | ± 35 V |
| P1-45 | P1-50 | 7:1 | 0.1% | 0 | 1 | X | X | 0 | 1 | ± 35 V |
| P1-54 | P1-49 | 7:1 | 0.1% | 0 | 1 | X | X | 1 | 0 | ± 35 V |
| P1-52 | P1-49 | 7:1 | 0.1% | 0 | 1 | X | X | 1 | 1 | ± 35 V |
| P1-9 | P1-1 | 3:1 | 0.1% | 1 | 0 | X | 0 | 0 | 0 | ± 15 V |
| P1-48 | P1-1 | 7:1 | 0.1% | 1 | 0 | X | 0 | 0 | 1 | ± 35 V |
| P1-44 | P1-1 | 7:1 | 0.1% | 1 | 0 | X | 0 | 1 | 0 | ± 35 V |
| P1-46 | P1-1 | 7:1 | 0.1% | 1 | 0 | X | 0 | 1 | 1 | ± 35 V |
| P1-42 | P1-1 | 7:1 | 0.1% | 1 | 0 | X | 1 | 0 | 0 | ± 35 V |
| P1-58 | P1-1 | 1:1 | - | 1 | 0 | X | 1 | 0 | 1 | ± 5 V |
| P1-62 | P1-1 | 1:1 | 2/ | 1 | 0 | X | 1 | 1 | 0 | ± 5 V |



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TABLE II. Input channel attenuation and select - Continued.

| (+) Input | (-) Input | Attenuation Ratio | Attenuation Accuracy | BUS 5 | 4 | 3 | 2 | 1 | 0 | Input voltage range |
|---|---|---|---|---|---|---|---|---|---|---|
| P1-60 | P1-1 | 1:1 | - | 1 | 0 | X | 1 | 1 | 1 | ± 5 V |
| P1-69 | P1-64 | 1:1 | - | 1 | 1 | 0 | 0 | X | X | ± 5 V |
| P1-39 | P1-50 | 7:1 | 0.1% | 1 | 1 | 0 | 1 | X | X | ± 35 V |
| P1-41 | P1-50 | 7:1 | 0.1% | 1 | 1 | 1 | 0 | X | X | ± 35 V |
| P1-63 | P1-57 | 1:1 | - | 0 | 0 | 1 | 1 | 0 | 1 | ± 5 V |

(X denotes a don't care condition.)

1/ Voltage at P1-83 = Voltage at P1-1, P1-82 is pulled up to +5 Vdc through a 4.7 kilohm (kohm)/resistor.

2/ These channels are intended for voltage division measurements. They are pulled up through precision 2 kohm resistors to the voltage present at P1 pins 9 and 10 (12 Vdc in normal use).

3.3.7 Signal processing.

3.3.7.1 Instrumentation amplifier. R3 shall be adjustable so that U10 can provide a differential gain of 1 ± 0.1% from U10-1 (+) and U10-3 (-) to U10-7 with a not less than 74 decibels (dB) CMRR, adjustable by R10, and an offset at U10-7 of not greater than 200 millivolts (mV) when the voltage at U10-1 and U10-3 is zero.

3.3.7.2 Summing amplifier. The ratio of the magnitude of the voltage at U12-14 with respect to U11-4 shall be 1 ± 0.1%. The ratio of the magnitude of the voltage at U12-14 with respect to U11-1 grounded minus the offset from the previous paragraph shall be not greater than 11 mV.

3.3.7.3 Gain select and accuracy. The gains shown in table III shall be selected when the data bits shown in table III are applied in accordance with the waveforms shown in figure 4. The ratio of the magnitude of the voltage at U13-13 with respect to U12-3 shall be as selected ± 0.098%. When the gain of 1 is selected and the voltage at U12-3 is zero, the voltage at U13-13 shall not exceed 10 mV.

TABLE III. Gain select.

| Gain | BUS 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| 1 | X | X | X | X | X | X | 0 | 0 |
| 10 | X | X | X | X | X | X | 0 | 1 |
| 30 | X | X | X | X | X | X | 1 | 0 |
| 60 | X | X | X | X | X | X | 1 | 1 |

(X denotes a don't care condition.)



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3.3.7.4 Filter select and attenuation. The -3 dB 2-pole low-pass filter frequencies shown in table IV shall be selected when the data bits shown are applied in accordance with the waveforms shown in figure 4. Each filter frequency selected shall conform to the attenuations shown in table V where fo is the selected -3 dB frequency.

TABLE IV. Filter select.

| -3 dB Freq. | BUS 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| 6 Hz 1/ | X | X | X | X | 0 | 0 | X | X |
| 60 Hz | X | X | X | X | 0 | 1 | X | X |
| 600 Hz | X | X | X | X | 1 | 0 | X | X |
| 6 kHz | X | X | X | X | 1 | 1 | X | X |

(X denotes a don't care condition.)
1/ Hz = hertz

TABLE V. Filter attenuation.

| Max. attenuation at (fo)/10 | Minimum attenuation at (fo) | 10(fo) |
|---|---|---|
| -0.1 dB | -2.36 dB | -26.4 dB |

3.3.8 ADCON. The ADCON (U15) shall begin converting the voltage present at U15-13 to a binary weighted word (12-bits) when a positive pulse, of no less than 100 ns in duration, is applied to U15-18. The conversion shall be complete 25 microseconds (µs) after the rising edge of the positive pulse applied to U15-18. The ADCON output shall conform to bit patterns (±2 LSB) in table VI for the voltages shown.

TABLE VI. ADCON conversion.

| Input voltage | ADCON MSB BUS 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | ADCON LSB BUS 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4.997 V | 0 | X | 0 | X | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4.500 V | 0 | X | 0 | X | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| 4.000 V | 0 | X | 0 | X | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 3.500 V | 0 | X | 0 | X | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3.000 V | 0 | X | 0 | X | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 2.500 V | 0 | X | 0 | X | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2.000 V | 0 | X | 0 | X | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1.500 V | 0 | X | 0 | X | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1.000 V | 0 | X | 0 | X | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0.500 V | 0 | X | 0 | X | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0.000 V | 0 | X | 0 | X | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| -0.500 V | 0 | X | 1 | X | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| -1.000 V | 0 | X | 1 | X | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| -1.500 V | 0 | X | 1 | X | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| -2.000 V | 0 | X | 1 | X | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| -2.500 V | 0 | X | 1 | X | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |



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TABLE VI. ADCON conversion - Continued.

| Input voltage | ADCON MSB BUS | | | | | | | | ADCON LSB BUS | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| -3.000 V | 0 | X | 1 | X | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
| -3.500 V | 0 | X | 1 | X | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| -4.000 V | 0 | X | 1 | X | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| -4.500 V | 0 | X | 1 | X | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| -4.999 V | 0 | X | 1 | X | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |

(X denotes a don’t care condition.)

3.3.9 Automated testing equipment (ATE) test. The PWA shall pass TACOM approved ATE test installed in TACOM approved ATE or equivalent.

3.3.10 Soldering. Soldering shall meet or exceed the requirements of ANSI/IPC J-STD-001 (see 4.2).

3.4 Interface requirements. Each PWA shall accommodate the following requirements.

3.4.1 Overall envelope. Overall envelope of the PWA shall be in accordance with drawing 12258777.

3.5 Ownership and support requirements.

3.5.1 Interchangeability. Interchangeability tolerances should permit parts, subassemblies and assemblies to be used in their parent assemblies without regard to the source of supply or manufacturer. Parts, subassemblies, and assemblies having the full range of dimensions and characteristics permitted by the specification governing the part, subassembly or assembly should be usable as replacement items without selection and without departure from the specified performance guidelines of the parent items.

3.5.2 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall include as a minimum the following:

a. Nomenclature “PRINTED WIRING ASSEMBLY, MUX/ADCON”.
b. Military part number.
c. Manufacturer’s cage code and name
d. Manufacturer’s serial number.
e. Contract number.
f. Symbol “US”.

3.6 Operating environment requirements. Each PWA shall operate under the following environmental conditions without damage or loss of performance.



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3.6.1 Temperature.

3.6.1.1 High temperature (operating). The PWA shall be capable of operation without damage or malfunction at a high temperature of +125 degrees Fahrenheit (°F) (52 degrees Celsius (°C)).(without solar radiation).

3.6.1.2 Low temperature (operating). The PWA shall be capable of operation without damage or malfunction at a low temperature of +20°F (-7°C). For ambient temperatures below +20°F (-7°C) , the test set shall be operated in a shelter.

3.6.1.3 High temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of +160°F (71°C).

3.6.1.4 Low temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of -60°F (-51°C).

3.6.2 Fungus. All nonmetallic materials that are not inherently fungistatic shall function in a fungus environment without adversely affecting the operation/performance of the PWA.

3.6.3 Shock. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the shock test.

3.6.4 Vibration. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the vibration test.

3.6.5 Burn-in. Unless otherwise specified (see 6.2), the PWA shall be subjected to and pass a “power on” burn-in prior to the inspections listed in 4.1.

4. VERIFICATION

4.1 Classification of inspection: The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (CI) (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on the initial production representative PWA of an order or when required by the contract. When a first article inspection is required (see 6.2 and 6.3), it includes all the verifications listed in table VII.

4.1.2 Conformance inspection. Conformance inspection shall include those examinations and tests from table VII on all PWA produced as specified in the contract.



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4.2 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

TABLE VII. Verification methods.

| Title | Requirement | Verification |
|---|---|---|
| Design, materials & manufacturing processes | 3.2 | 4.2 |
| Operating requirements | 3.3 | 4.2.3 |
| Input power | 3.3.1 | 4.2.3.1 |
| Logic levels | 3.3.2 | 4.2.3.2 |
| Output clocks | 3.3.3.1 | 4.2.3.3.1 |
| Output latches | 3.3.3.2 | 4.2.3.3.2 |
| Voltage level shifters | 3.3.3.3 | 4.2.3.3.3 |
| Input clocks | 3.3.3.4 | 4.2.3.3.4 |
| Input buffers | 3.3.3.5 | 4.2.3.3.5 |
| Internal logic levels | 3.3.4 | 4.2.3.4 |
| Input channel max. select | 3.3.5 | 4.2.3.5 |
| Input channel attenuation | 3.3.6 | 4.2.3.6 |
| Instrumentation amplifier | 3.3.7.1 | 4.2.3.7.1 |
| Summing amplifier | 3.3.7.2 | 4.2.3.7.2 |
| Gain select & accuracy | 3.3.7.3 | 4.2.3.7.3 |
| Filter select & attenuation | 3.3.74 | 4.2.3.7.4 |
| ADCON | 3.3.8 | 4.2.3.8 |
| ATE test 1/ | 3.3.9 | 4.2.3.9 |
| Interface requirements | 3.4 | 4.2.4 |
| PWA overall envelope | 3.4.1 | 4.2.4.1 |
| Ownership & support requirements | 3.5 | 4.2.5 |
| Interchangeability | 3.5.1 | 4.2.5.1 |
| Identification marking | 3.5.2 | 4.2.5.2 |
| Operating environment requirements | 3.6 | 4.2.6 |
| Temperature | 3.6.1 | 4.2.6.1 |
| Fungus | 3.6.2 | 4.2.6.2 |
| Shock | 3.6.3 | 4.2.6.3 |
| Vibration | 3.6.4 | 4.2.6.4 |

1/ Not required unless specified (see 6.2).



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4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25 ± 8°C).
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.2.3 Operating requirements verification.

4.2.3.1 Input power. To determine conformance to 3.3.1, the input current shall be measured with an ammeter.

4.2.3.2 CMOS Logic levels. To determine conformance to 3.3.2, the input and output CMOS logic levels shall be observed and verified with an oscilloscope.

4.2.3.3 Input/output logic.

4.2.3.3.1 Output clocks. To determine conformance to 3.3.3.1, a logic analyzer shall be used to verify that the signal at U7-10 conforms to the waveforms shown in figure 1. Using a logic analyzer, the signal at U7-4 shall be verified to conform to the waveforms shown in figure 2.

4.2.3.3.2 Output latches. To determine conformance to 3.3.3.2, a logic analyzer shall be used to verify that the logic levels applied to BUS 0 through BUS 5 (P1-21 through P1-26) appear at the outputs specified when signals conforming to those shown in figures 3 and 4 are applied to the PWA.

4.2.3.3.3 Voltage level shifters. To determine conformance to 3.3.3.3, a logic analyzer and a multimeter shall be used to verify signal timing and voltage levels.

4.2.3.3.4 Input clocks. To determine conformance to 3.3.3.4, a logic analyzer shall be used to verify that the signal at U7-3 conforms to the waveforms shown in figure 5. Using a logic analyzer, the signal at U7-11 shall be verified to conform to the waveforms shown in figure 6.

4.2.3.3.5 Input buffers. To determine conformance to 3.3.3.5, a logic analyzer shall be used to verify that logic levels at the points specified drive BUS 0 through BUS 7 (P1-21 through P1-28) when signals conforming to those shown in figures 7 and 8 are applied to the PWA. The logic level at U6-11 shall be verified to conform to the requirements and time specified.

4.2.3.4 Internal logic levels. Conformance to 3.3.4 shall be demonstrated by successful completion of the tests specified in paragraphs 4.2.3.5, 4.2.3.7.3 and 4.2.3.7.4.



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4.2.3.5 Input channel Mux select. To determine conformance to 3.3.5, for each (+, -) input pair specified in table II, the corresponding bit pattern shall be applied to BUS 0 through BUS 5 (P1-21 through P1-26) and control signals conforming to the waveforms shown in figure 3. A dc voltage (within the input voltage range specified) shall be applied to the specified input and the input signal, times the attenuation ratio, shall be verified to be present at U10 pins 1 (+) and 3 (-).

4.2.3.6 Input channel attenuation. To determine conformance to 3.3.6, the channels listed in table II that have an attention ratio greater than 1:1 shall be selected. For each, the attenuation ratio shall be verified to be within the tolerance shown in table II using the procedure in paragraph 4.2.3.5.

4.2.3.7 Signal processing.

4.2.3.7.1 Instrumentation amplifier. To determine conformance to 3.3.7.1, the tests of 4.2.3.7.1.1 and 4.2.3.7.1.2 shall be performed.

4.2.3.7.1.1 Gain adjust. Select the first Mux channel listed in table II. Select a gain of 1 as listed in table III. Select filter -3dB frequency of 6 kHz as listed in table IV. Ground pins P1-65 and P1-67. Apply a variable voltage dc source to P1-68 and adjust the voltage until the voltage at U12-7 is 0.000 Vdc. Remove P1-65 and P1-67 from ground. Apply 3.5 Vrms ac at 100 Hz to P1-65 (+) and P1-67 (-). Measure the voltage between P1-65 and P1-67 and note the value. Adjust R3 (gain adjust) until the ac voltage at U12-7 is equal to the previously noted value.

4.2.3.7.1.2 CMRR adjust. Select the input channel P1-55/P1-53 as listed in table II. Ground pins P1-55 and P1-53. Select gain of 60 as listed in table III. Select a filter -3dB frequency of 6 kHz as listed in table IV. Apply a variable voltage dc source to P1-68 and adjust for 0.000 Vdc at U12-7. Disconnect P1-55 and P1-53 from ground. Apply 14.14 Volts root mean square (Vrms) alternating current (ac) at 100 Hz to P1-55/P1-53 (both connected to plus), and a ground. Adjust R10 for minimum ac voltage at U12-7. Verify that the ac voltage at U12-7. Verify that the ac voltage after adjustment is no greater than 8.4 millivolts root mean square (mVrms).

4.2.3.7.2 Summing amplifier. To determine conformance to 3.3.7.2, the following test shall be performed. Select the first Mux channel as listed in table II. Ground P1-67 and P1-68. Apply a variable voltage dc supply to P1-65. Adjust the voltage at P1-65 until the voltage at U10-7 is 0.000 Vdc. Verify that the summing amplifier offset voltage at U12-14 is not more than +11 mVdc. Adjust the voltage at P1-65 until the voltage at U10-7 is +5.000 Vdc. Verify that the voltage at U12-14 is +5.000 $\pm$ 0.01 Vdc when the previously measured summing amplifier offset voltage is subtracted. Remove P1-68 from ground. Apply +5.000 Vdc to P1-68. Adjust the voltage at P1-65 until the voltage at U10-7 is 0.000 Vdc. Verify that the voltage is U12-14 is -5.000 $\pm$ 0.005 Vdc after the summing amplifier offset voltage is subtracted.



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4.2.3.7.3 Gain select and accuracy. To determine conformance to 3.3.7.3, the following test shall be performed. Select the first Mux channel listed in table II. Adjust the voltage at P1-68 until the voltage at U12-3 is zero. Apply the bit pattern for the gain of 1 from table III to BUS 0 through BUS 7 (P1-21 through P1-28) along with the required control signals in conformance with figure 4. Verify that the offset voltage at U13-13 is not greater than 10 mV. Adjust the voltage at P1-68 until the voltage at U13-13 is zero. Apply +5.000 Vdc to U10-1. Verify that the voltage at U13-13 is 5.000 $\pm$ 0.040 Vdc. Select the gain of 10. Apply 0.240 Vdc to U10-1. Verify that the voltage at U13-13 is 2.500 $\pm$ 0.040 Vdc. Select the gain of 30. Apply 0.100 Vdc to U10-1. Verify that the voltage at U13-13 is 3.000 $\pm$ 0.024 Vdc. Select the gain of 80. Apply 0.070 Vdc to U10-1. Verify that the voltage at U13-13 is 4.200 $\pm$ 0.034 Vdc.

4.2.3.7.4 Filter select and attenuation. To determine conformance to 3.3.7.4, the following test shall be performed. Select the first Mux channel listed in table II. For each filter frequency listed in table IV, apply the bit patterns (with BUS 1 and BUS 0 set to zero for a gain of 1) to BUS 0 through BUS 7 (P1-21 through P1-28) along with the required control signals in conformance with figure 4. Apply an 8.000 V p-p AC waveform across U10-1 and U10-3 and verify the attenuations at each frequency in table V. Note that any input below 5 Hz can be considered a dc voltage.

4.2.3.8 ADCON. To determine conformance to 3.3.8, perform the following procedure for each of the input voltages listed in table VI. Apply the dc input voltage to P1-72. Apply a positive going logic pulse of not less than 100 ns nor more than 1.3 µs duration to P1-40. 25 µs after application of the logic pulse to P1-40, apply the waveforms shown in figure 7 and verify that the bit pattern on BUS 7 through BUS 0 conform to those in table VI for the ADCON MSB. Apply the waveforms shown in figure 8 and verify that the bit pattern on BUS 7 through BUS 0 conforms to those in table VI $\pm$ 2 LSB for the ADCON LSB.

4.2.3.9 ATE Test. To determine conformance to 3.3.9, install the PWA into TACOM approved fixture (such as BTF-2500-02) on TACOM approved ATE equipment (such as HP3062 with TACOM approved software and instructions (such as BSW-25000020).

4.2.4 Interface requirements verification.

4.2.4.1 PWA overall envelope inspection. Insert the PWA into the present existing location of the test set, and verify its envelope and interfacing conformance.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Interchangeability inspection. To determine conformance to 3.5.1, the interchangeability between different part, subassemblies, and assemblies having the same military part number shall be verified.



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4.2.5.2 Identification marking. Verify the presence of the required PWA identification marking.

4.2.6 Operating environment requirements verification. The electrical performance tests specified in 4.2.3.1 through 4.2.3.8 shall be conducted prior to and at the completion of each environmental test. Verification of performance during high temperature and low temperature testing shall be demonstrated by the tests specified in 4.2.3.3 through 4.2.3.8.

4.2.6.1 Temperature. To determine conformance to 3.6.1, the PWA shall be tested as follows.

4.2.6.1.1 High temperature (operating). To determine conformance to 3.6.1.1, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure II (hot category), of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.2 Low temperature (operating). To determine conformance to 3.6.1.2, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure II, of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.3 High temperature (nonoperating). To determine conformance to 3.6.1.3, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.1.4 Low temperature (nonoperating). To determine conformance to 3.6.1.4, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.2 Fungus. To determine conformance to 3.6.2, the PWA shall be subjected to the applicable fungus test in accordance with method 508.4 of MIL-STD-810 (28 day test), or equivalent (see 4.2.1).

4.2.6.3 Shock. To determine conformance to 3.6.3. The PWA shall be subjected to the shock test in accordance with method 516.4, Procedure I, 40g peak of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3 after shock test.

4.2.6.4 Vibration. To determine conformance to 3.6.4, the PWA shall be subjected to a vibration test using a composite of figures 9 through 11. These figures represent the cargo environment at the cargo bed of a composite of tactical wheeled vehicles, the 5-ton M813 truck and the 12-ton M127 semi- trailer. The data include differing vehicle loading conditions traversing over specially designed courses ranging from paved highway to off road conditions at various vehicle speeds. The spectrum is broadband random with peaks and notches at various discrete frequency bands. Break points are provided for establishing the spectrum shape. The test duration is 500 miles. The PWA shall be tested to 3.3 after the vibration test.



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4.2.6.5 Burn-in. When required (see 6.2), the burn-in shall consist of three temperature cycles and a six gravity unit (g) root mean square (rms) random vibration as follows:

a. Temperature cycle. Each temperature cycle shall consist of 30 minutes at minus (-) 60°F (-51°C) and 30 minutes at plus (+) 125°F (52°C). The rate of temperature change between temperature extremes shall be not less than 5°F (2.8°C) per minute. Power shall be applied to the PWA for not less than 30 minutes from 20°F (7°C) through the completion of each high-temperature soak.
b. Random vibration. Following all temperature cycling, the PWA shall be hard mounted to the vibration table so that the axis of vibration is perpendicular to the plane of the PWA. The vibration shall be at room ambient temperature, for a period of 10 minutes, as shown in figure 12.

NOTE: PWAs which will be procured for simultaneous installation in VTMs being procured do not have to be subjected to “burn-in” provided the VTMs are subjected to the 16-hour “power on” burn-in.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency , or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The PWA multiplexes nineteen 5 V channels, one 15 V channel, eleven 35 V channels, and one 50 V channel to a single high- impedance instrumentation amplifier. A summing amplifier allows an offset to be subtracted from the output of the instrumentation amplifier. Gains of 1, 10, 30, and 60 along with 2-pole low-pass filter frequencies of 6, 60, 600, and 6000 Hz are software selectable. A 12-bit ADCON (analog to digital converter) with a $\pm$ 5 V signal range can convert an analog input to a digital output every 25 microseconds. The PWA is



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military unique because it is designed specifically to be a component of the Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM), and to withstand environmental conditions which exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. If first article inspection is required (see 3.1).
d. PWA drawing numbers, revision letter, date and PWA part number (see 3.2 and 3.5.2).
e. If marking is other than as specified (see 3.5.2).
f. If burn-in is not required (see 3.6.5).
g. If CMOS logic levels should be other than as specified (see 3.3.2).
h. Whether ATE test is required (see table VII footnote).
i. Packaging requirements.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table VII and their designated frequency based on a risk assessment for the procurement.

6.5 Subject term (key word) listing.

Analog to digital convertor (ADCON)
Ignition inhibits
Ignition signal outputs
Multiplex (Mux)
Simplified test equipment/Internal combustion engine - Reprogrammable (STE/ICE-R)



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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

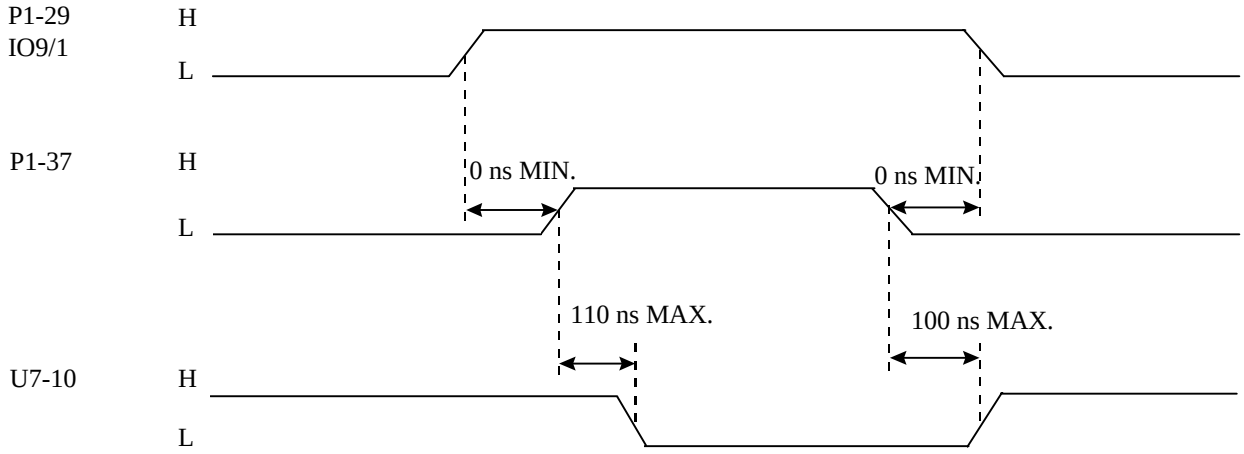



FIGURE 1. Waveforms to generate U7-10 clock.

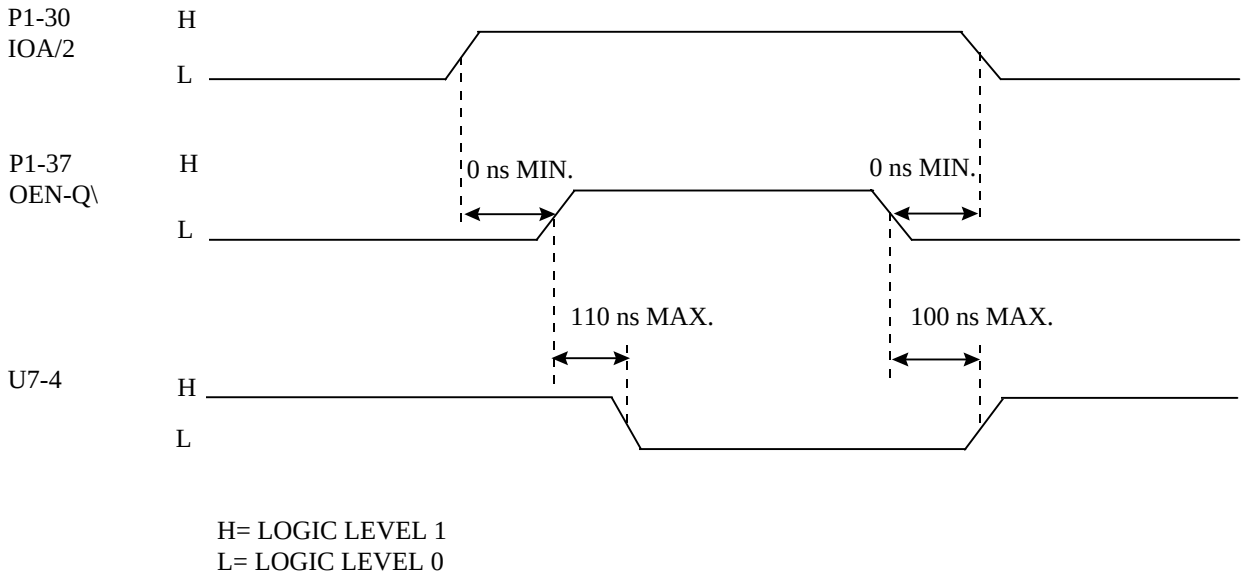
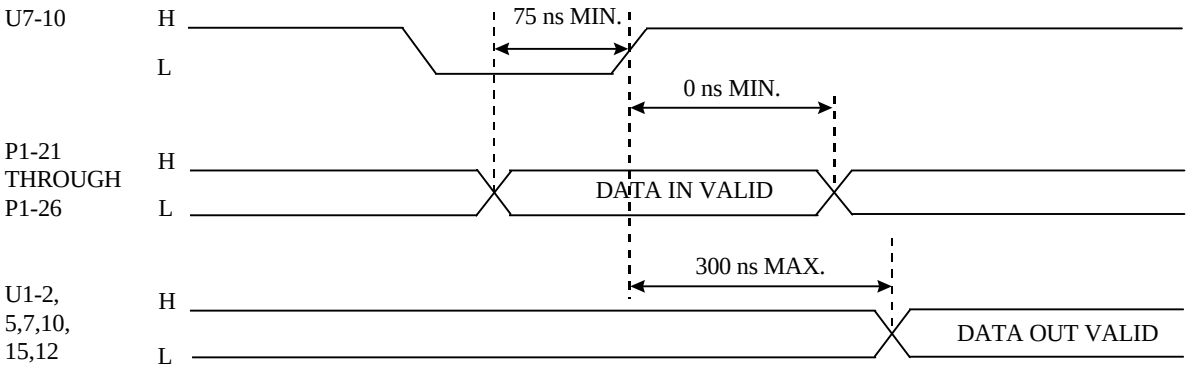



FIGURE 2. Waveforms to generate U7-4 clock.

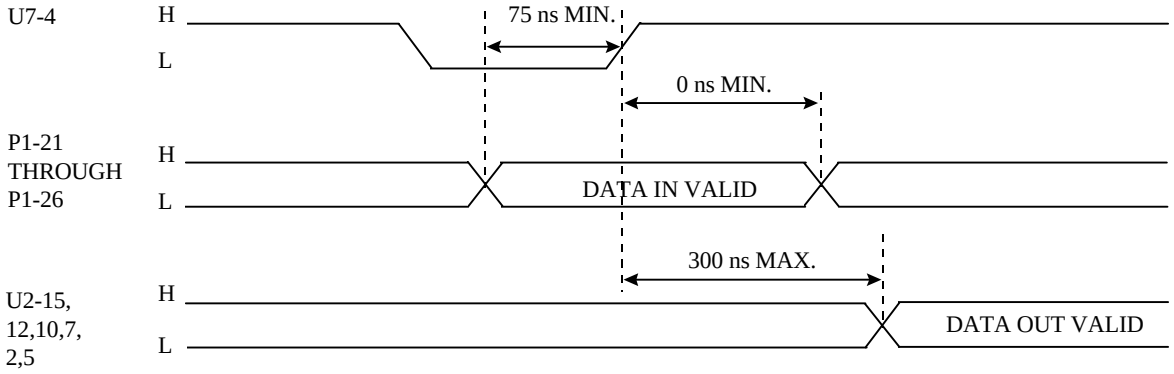


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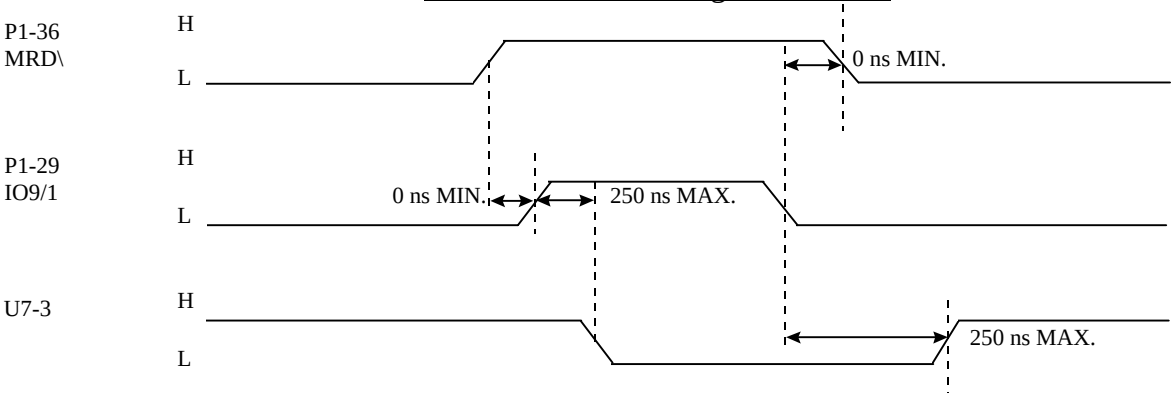

H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 3. Waveforms for latching data into U1.




H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 4. Waveforms for latching data into U2.




H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 5. Waveforms to generate U7-3 clock.



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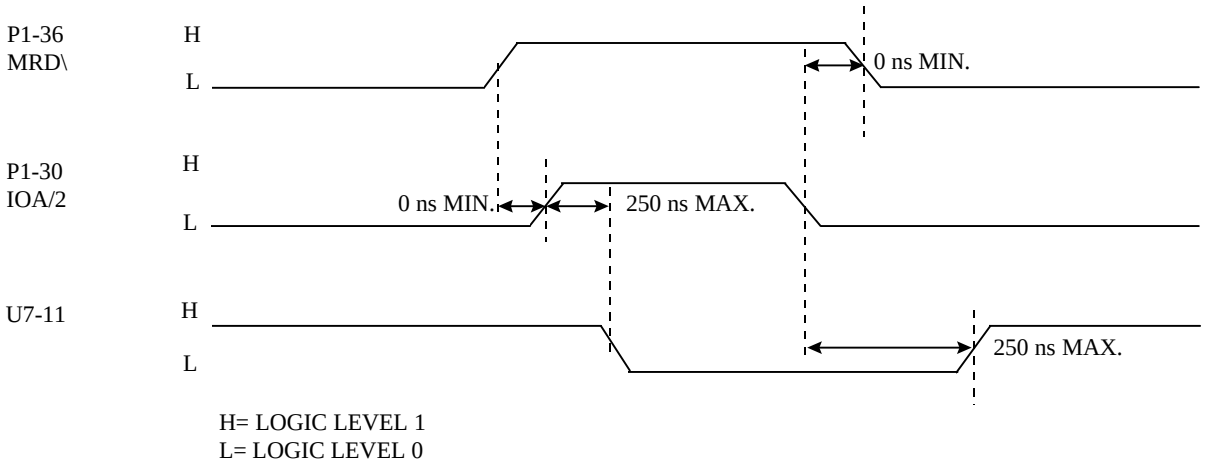



FIGURE 6. Waveforms to generate U7-11 clock.

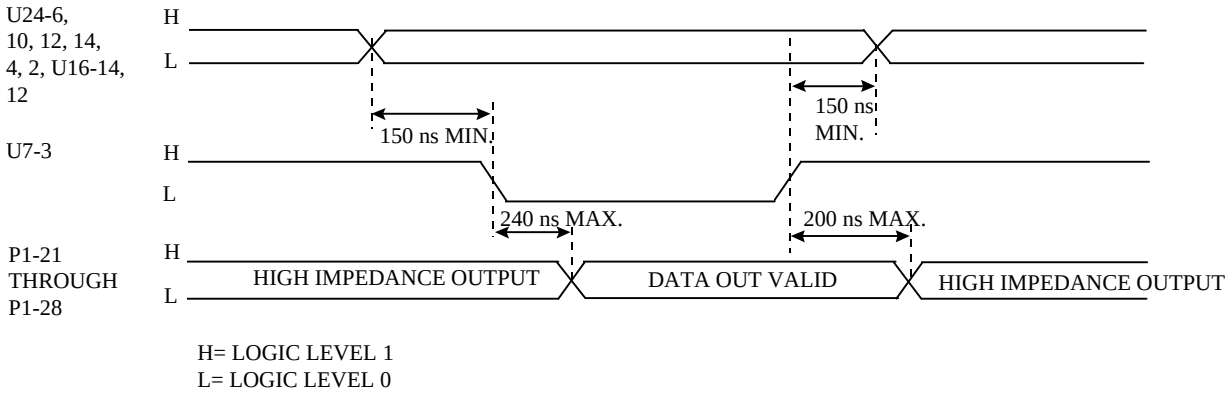



FIGURE 7. Waveforms for reading U24 and U16 buffers.

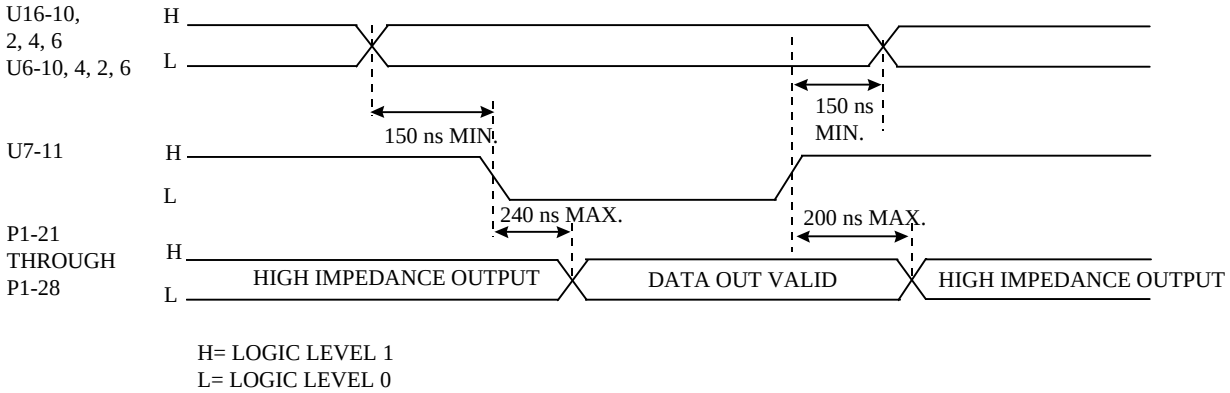
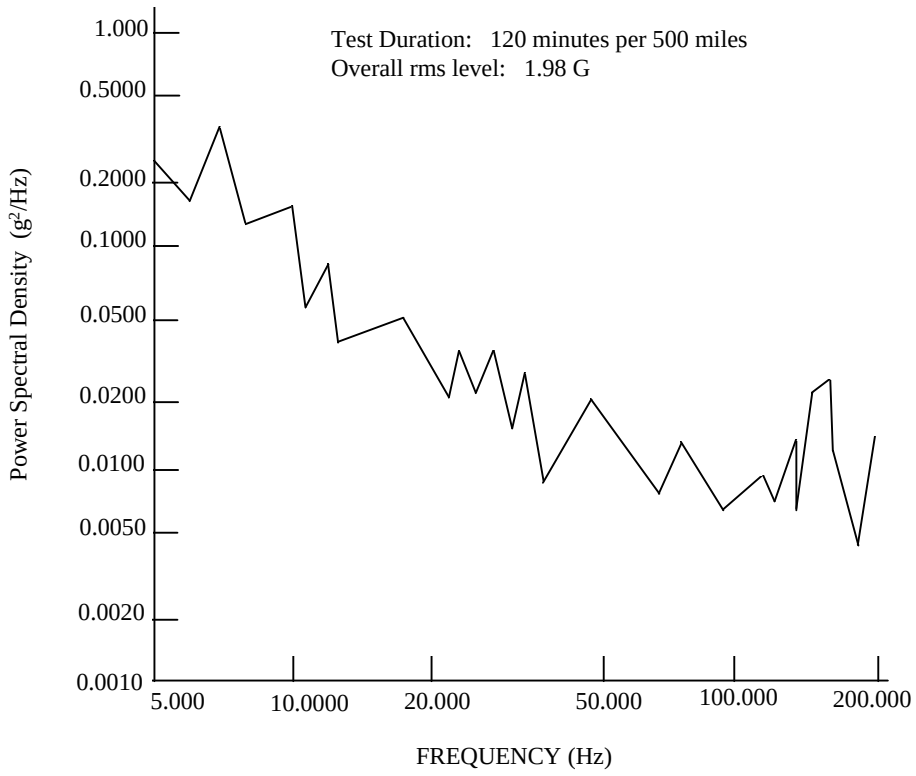



FIGURE 8. Waveforms for reading U16 and U6 buffers.



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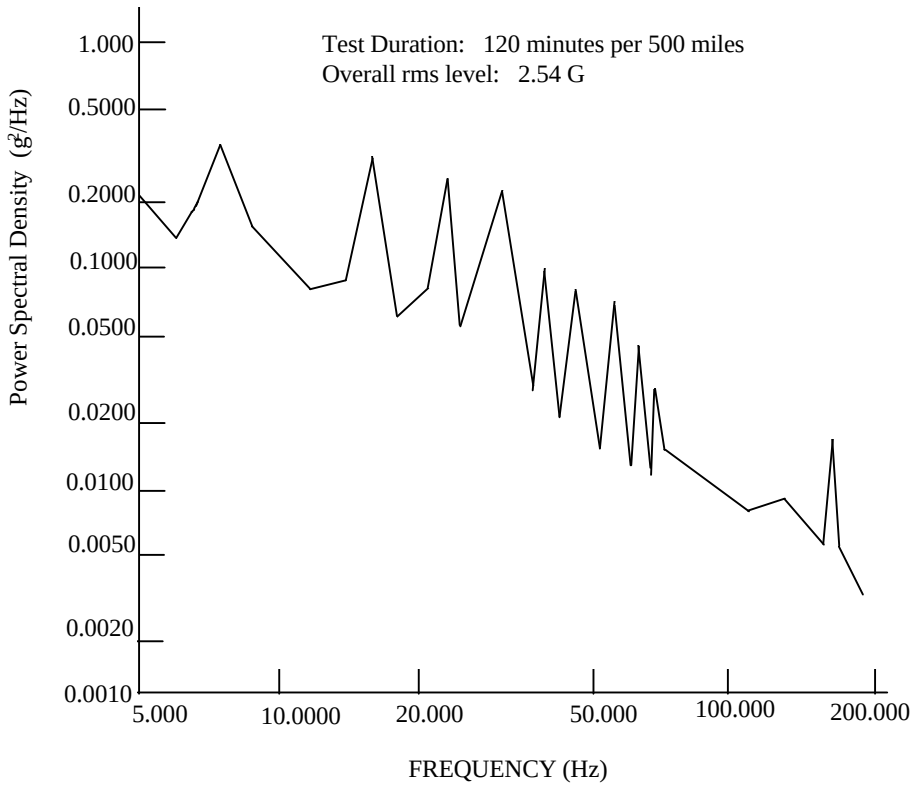



| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.23840 |
| 6 | 0.16840 |
| 7 | 0.38230 |
| 8 | 0.14870 |
| 10 | 0.16840 |
| 11 | 0.05930 |
| 12 | 0.09750 |
| 13 | 0.04090 |
| 18 | 0.05370 |
| 23 | 0.01940 |
| 24 | 0.03520 |
| 26 | 0.02310 |
| 28 | 0.03520 |
| 32 | 0.01670 |
| 33 | 0.03110 |
| 37 | 0.00810 |
| 49 | 0.02190 |
| 63 | 0.00700 |
| 73 | 0.01150 |
| 98 | 0.00660 |
| 120 | 0.00970 |
| 123 | 0.00680 |
| 138 | 0.01340 |
| 140 | 0.00760 |
| 146 | 0.02490 |
| 159 | 0.02820 |
| 163 | 0.01300 |
| 188 | 0.00480 |
| 200 | 0.01340 |

FIGURE 9. Basic transportation, composite tactical wheeled environment, vertical axis.



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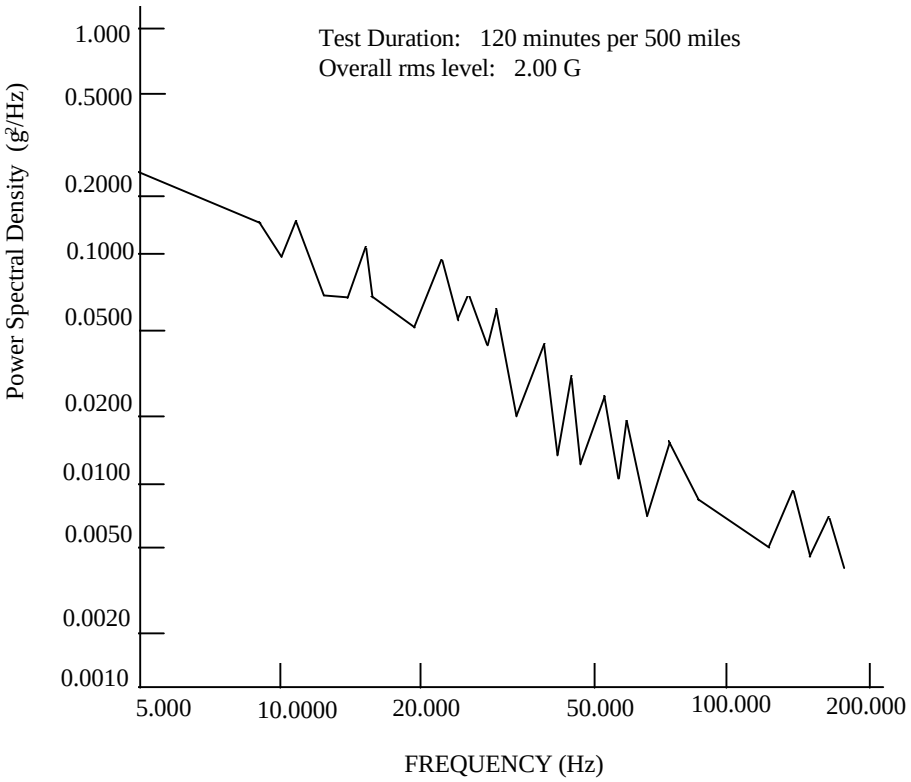



| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.22130 |
| 6 | 0.13140 |
| 7 | 0.20540 |
| 8 | 0.38230 |
| 9 | 0.15630 |
| 12 | 0.07990 |
| 15 | 0.08830 |
| 16 | 0.32940 |
| 19 | 0.06080 |
| 22 | 0.09280 |
| 24 | 0.26340 |
| 27 | 0.05640 |
| 32 | 0.22690 |
| 36 | 0.02820 |
| 40 | 0.11600 |
| 44 | 0.02090 |
| 48 | 0.08190 |
| 53 | 0.01480 |
| 56 | 0.07420 |
| 60 | 0.01270 |
| 64 | 0.04980 |
| 65 | 0.01100 |
| 68 | 0.03260 |
| 69 | 0.01480 |
| 114 | 0.00740 |
| 132 | 0.00830 |
| 160 | 0.00510 |
| 169 | 0.01710 |
| 173 | 0.00550 |
| 200 | 0.00310 |

FIGURE 10. Basic transportation, composite tactical wheeled environment, longitudinal axis.



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| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.25490 |
| 10 | 0.13030 |
| 11 | 0.09670 |
| 12 | 0.13360 |
| 13 | 0.06500 |
| 15 | 0.06340 |
| 16 | 0.11230 |
| 17 | 0.06500 |
| 22 | 0.04590 |
| 24 | 0.08980 |
| 26 | 0.05070 |
| 28 | 0.06830 |
| 30 | 0.03670 |
| 32 | 0.05880 |
| 38 | 0.01920 |
| 40 | 0.03860 |
| 44 | 0.01290 |
| 48 | 0.02940 |
| 50 | 0.01140 |
| 56 | 0.02460 |
| 60 | 0.00960 |
| 63 | 0.01880 |
| 70 | 0.00660 |
| 78 | 0.01500 |
| 93 | 0.00730 |
| 137 | 0.00460 |
| 145 | 0.00910 |
| 162 | 0.00430 |
| 176 | 0.00660 |
| 200 | 0.00360 |

FIGURE 11. Basic transportation, composite tactical wheeled environment, transverse axis.



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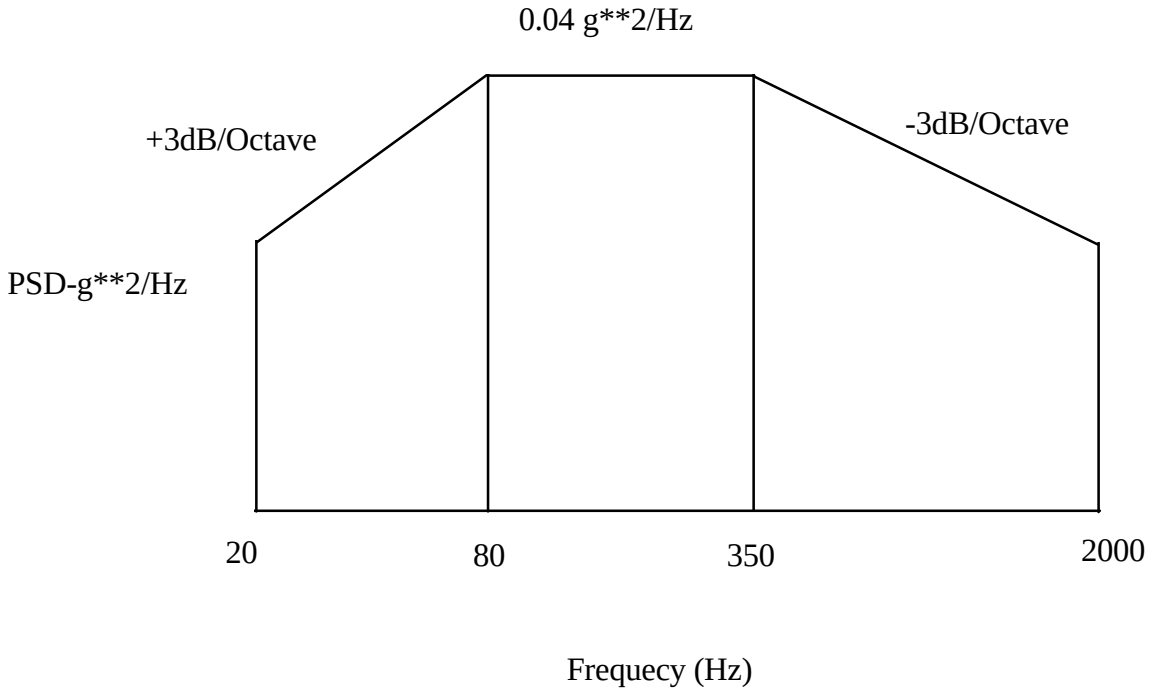



FIGURE 12. Random vibration spectrum.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 5998-0076)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-62741A(AT) | 2. DOCUMENT DATE *(YYMMDD)*<br>980123 |
|---|---|---|

**3. DOCUMENT TITLE**
PRINTED WIRING ASSEMBLY, MUX/ADCON

**4. NATURE OF CHANGE** ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON<br>*(If applicable)* | 7. DATE SUBMITTED<br>*(YYMMDD)* |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial (2) AUTOVON |
|---|---|
| c. ADDRESS *(Include Zip Code)*<br>Commander<br>U.S. Army Tank-automotive and Armaments Command<br>ATTN: AMSTA-TR-E/BLUE<br>Warren, MI 48397-5000 | (810) 574-8745 786-8745<br>IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403<br>Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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